

YAD! (Yet Another Drummer)

YAD! (YET ANOTHER DRUMMER!) V.1.1.0

User Manual



YAD! (also Yet Another Drummer!) is a sample player designed primarily (but not exclusively) for drums, coded in C++ with Juce Framework version 8.0.12.

1. The Interface

- **The top section:** with the preset management buttons on the right (Load and Save Kit buttons) and the buttons "Options" (this open Juce menu for audio-midi setting) and the "Extra Map" button on the left.



- **The Drum Kit View:** the widest central section features an interactive visual representation of a drum kit. The pads are eight semi-transparent circles with a dot on the edge of the circumference. You can trigger sounds by clicking the pads or using your MIDI keyboard (mapped to standard GM Drum MIDI notes).



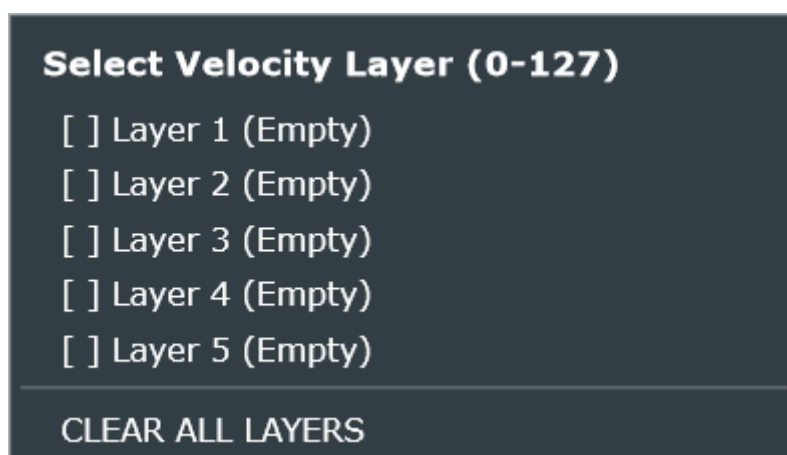
- **The Mixer (Bottom Panel):** this area includes the elements below
 - **OUT [1-8]:** Direct output routing selector.
 - **LOAD:** Sample and layer management for any drum element.
 - **Gain Sliders:** Horizontal faders to balance the volume of each drum element.



The Mixer (Bottom Panel)

2. Creating Your Library (Multi-Layer System)

Unlike basic samplers, YAD! supports up to **5 velocity layers** per drum pad. This allows for a "human" feel where different samples are triggered based on how hard you hit the key. By clicking the **"LOAD [DrumName]"** button, users can access a context menu to load samples into specific velocity layers (1-5).



2.1 How to Load Samples:

1. Click the **LOAD** button on a specific channel (e.g., Kick).
2. A menu will appear asking you to choose a **Velocity Layer (1-5)**.
3. Select a file (.wav or .aiff). We recommend a one-shot sample corresponding to the sound of the chosen drum instrument, with increasing intensity for each of the five layers..
4. **Automatic Mapping:** The engine automatically redistributes the MIDI velocity range (0-127).
 - If you load **1 sample**, it plays at all velocities.
 - If you load **3 samples**, the engine splits the range into roughly 0-42, 43-85, and 86-127.
5. The layer already occupied by a loaded sample is indicated with a check mark on the left. If you load even just one sample on a single layer, the red dot on the pad turns green.
6. **Clear All Layers:** This command has been added from version 1.1 to the "LOAD" menu to instantly wipe all samples from the selected pad, resetting it to an empty state without affecting other pads.

Select Velocity Layer (0-127)

[X] Layer 1 (LOADED)

[X] Layer 2 (LOADED)

[] Layer 3 (Empty)

[] Layer 4 (Empty)

[] Layer 5 (Empty)

CLEAR ALL LAYERS

Please Note: When a sample is loaded onto a layer, the dot at the edge of the pad's circumference changes color from **red** (indicating no samples on all layers) to **green** (indicating a sample is loaded onto at least one layer; note: at least one sample even if only on a single layer does green the dot on the pad).

Pro Tip for Realism:

To create a professional-sounding kit, load a "Soft" hit in Layer 1, a "Medium" hit in Layer 3, and a "Hard/Power" hit in Layer 5. The engine will handle the cross-fading and selection logic seamlessly.

2.2 THE DRUM MIDI MAPPING ON PADS:

The midi mapping of the pads or drum instruments in the main view follows the General Midi standard as shown in the table below:

CURRENT MIDI MAPPING OF THE PADS

Pad	MIDI Note	Note Number	Keyboard key (Standard)
KICK	C1	36	Do 1
SNARE	D1	38	Re 1
HI-HAT	F#1	42	Fa# 1
CRASH	C#2	49	Do# 2
TOM 1	A1	45	La 1
TOM 2	B1	47	Si 1
TOM 3	C2	48	Do 2
RIDE	D#2	51	Re# 2

3. EXTRA MAP

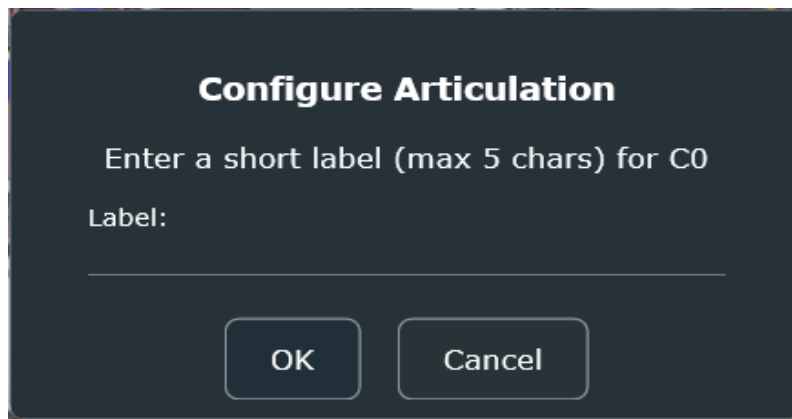


By clicking on the "Extra Map" button at the top left, a popup menu opens to load other samples on the midi notes not occupied by the 8 pads in the main view (these 8 pads are Kick, Hi-Hat, Crash, Snare, three Toms, Ride). The extra-maps are configurable by the user who can select the chosen notes among those distributed over five octaves, assign it a name or an acronym (a maximum of 5 characters of text or number are allowed) and then load the chosen sample for each layer, up to a maximum of three layers. These are fully integrated into the MIDI engine. It is recommended to use these additional maps to load samples of extra articulations ("Drum Rudiments") such as fills, flams, paradiddles, double-strokes, rolls, etc... but you can load also other one-shot samples of additional drum instruments (or other completely different sounds)

Custom sounds mapped via the **Extra Map** are via midi triggered (on previous version 1.0.0 they was only just visual placeholders).

- Each note map supports up to three layers (one sample per layer), as opposed to the main elements which support 5.
- Any MIDI Note-On message received on a mapped note (within Octaves 1-5) will trigger the corresponding articulation.
- **Reserved Notes:** The system automatically detects notes already assigned to the 8 standard pads (Kick, Snare, etc.) and marks them as **[RESERVED]** in the Extra Map to prevent MIDI conflicts.

Please note: as already mentioned above, before loading samples into the "Extra Maps" you must assign a label or acronym (up to a maximum of 5 characters of text and/or numbers, e.g. "Fill", "FLAM", "Roll", "") that helps memorize the correspondence between the selected MIDI note and the sound you want to play. After assigning the name you must return to the "Extra Map" menu, return to the layer and load the chosen sample. For each mapped note you have three layers, so you can load up to a maximum of three samples, always following an order of increasing intensity.



Configure Articulation

Enter a short label (max 5 chars) for C0

Label:

OK Cancel

Notes already assigned to items in the main view appear grayed out as already committed and therefore cannot be used in Extra Maps

```

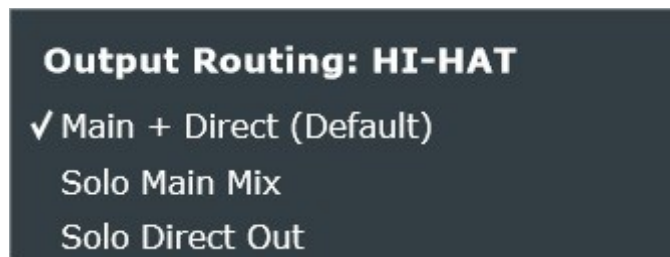
C1: [RESERVED FOR KICK]
C#1: [Empty]
D1: [RESERVED FOR SNARE]
D#1: [Empty]
E1: [Empty]
F1: [Empty]
F#1: [RESERVED FOR HI-HAT]
G1: [Empty]
G#1: [Empty]
A1: [RESERVED FOR TOM 1]
A#1: [Empty]
B1: [RESERVED FOR TOM 2]

```

4. Managing Kits (Save & Load)

- **SAVE KIT:** Click this in the top right to export your entire setup as a .xzkit file. This file remembers the paths to all 40 possible sample slots (8 pads x 5 layers) and your gain settings.
- **LOAD KIT:** Use this to recall a previously saved .yadkit.
 - *Note: Ensure your audio samples remain in the same folder location on your hard drive, as the kit file stores file paths.*
 - *The .yadkit preset looks for samples in the "Samples" folder which must be placed in the same location as the preset according to this directory structure: ,,,\yadkit folder\Samples*

5. Audio Routing



Each drum element can be routed to a specific output in your DAW. Click the **OUT** button above the load button to toggle between:

- **Main:** Stereo Master Mix.
- **Direct:** Individual mono/stereo outputs for separate processing (EQ, Dynamic Compression, etc) in your DAW mixer.

6. Extra Map: Multi-Output Routing



The **Extra Map** configuration now allows for independent output routing for each custom articulation.

- In the Extra Map menu, under each mapped note, you will find an **"Output Routing"** submenu.
- You can route specific articulations to the **Main Mix** or to one of the **4 Extra Stereo Groups**, allowing for separate processing (e.g., EQ or Compression) in your DAW.

7. SAMPLES REQUIREMENTS

7.1. Supported Formats

Current code (via `juce::AudioFormatManager`), supports exactly what we recorded in the constructor: `.wav`, `.aif` / `.aiff`, `.flac`.

7.2. Sample Rate and Bit Depth

The engine is extremely flexible because JUCE **resamples on the fly** if necessary:

- Sample Rate: Supports any rate (**44.1, 48, 88.2, 96, 192 kHz**). If the sample rate is 96kHz and your sound card is 44.1kHz, the plugin will still read the file correctly.
- Bit Depth: Supports **16-bit, 24-bit, and 32-bit (both integer and floating point)**. Internally, the plugin converts everything to 32-bit float for audio processing, ensuring the highest possible fidelity.

7.3. File Naming

No limitations. The system isn't based on the file name (like some samplers that look for "velocity" in the name), but on the location where you choose to load it via the interface. You can name the files whatever you like.

7.4. Hybrid Instruments and Experimentation

Absolutely! The plugin doesn't "know" that the sound is a snare drum.

- You can load a bass drum on Layer 1 and a human scream on Layer 5: at low velocity, you'll hear the bass; if you hit hard, you'll hear the scream.
- You can create hybrid kits (e.g., real kick drum, electronic snare drum, reverse-recorded cymbals). It's the perfect tool for creative sound design.

7.5. File Length

Technically, there's no limit (you could load a 10-minute file), but remember that:

- The plugin loads the entire sample into RAM.
- If you load 40 very long samples (8 pads x 5 layers), memory usage will increase dramatically. For a drum kit (single hits), this is never a problem.

7.6. Loop Support *(not recommended)*

Here we need to make an important technical distinction:

- Loading: You can load a file containing a loop.
- Playback: The plugin is currently a One-Shot Sampler. When it receives a MIDI note, it plays the file from start to finish (or until the sample ends), therefore by clicking on a midi note on which the loop is loaded, the latter will play in its entirety from beginning to end without any possibility of control either on a portion of the loop (slice) or on the bpm
- Current Limitations: We haven't implemented the "Loop Point" (i.e., the part that repeats as long as you hold down the key). If you load a 2-bar drum loop, it will only play it all once. No slicing system.
- Multi-layer: Yes, you can load different loops on different layers of the same pad (e.g., a calm shaker loop at low velocity and a frenetic one at high velocity).

8. PAD MAP AND EXTRA MAP

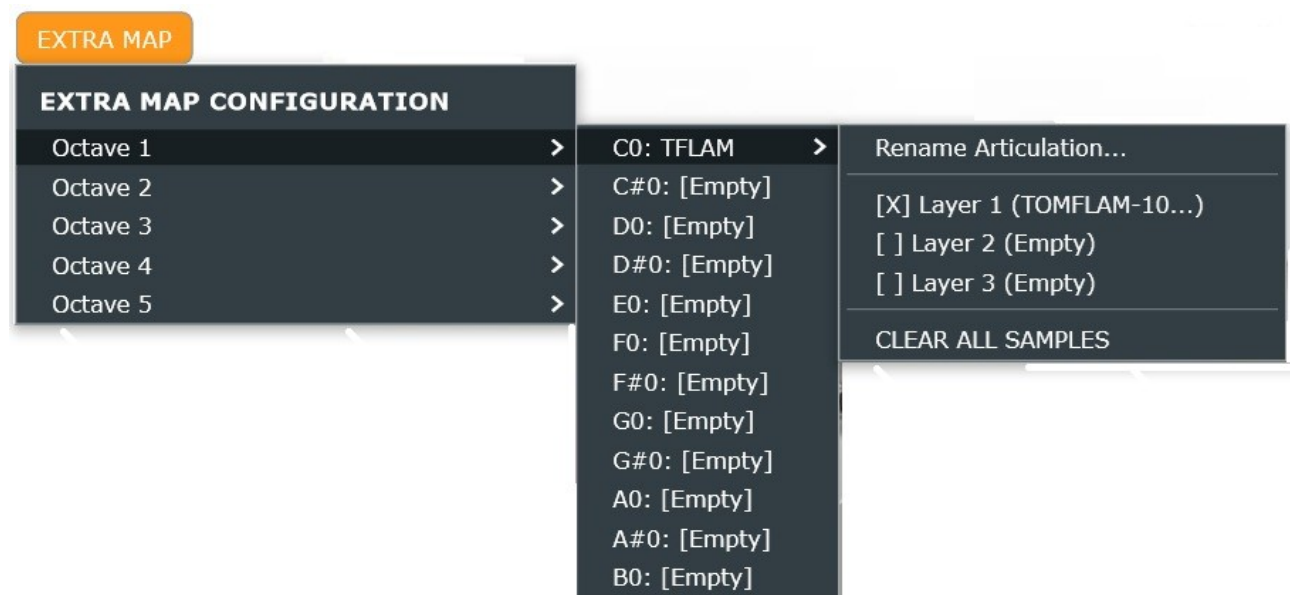
8.1. The Multi-Layer Sampling System

Each of the 8 Standard Pads (Kick, Snare, etc.) supports up to 5 **velocity layers**.

- **How to load:** Click the "**Load**" button on any pad. A menu will appear showing 5 slots.
- **Visual Feedback:**
 - ☐ Layer X (Empty): No sample loaded.
 - ☒ Layer X (LOADED): A sample is active.
- **Velocity Logic:** The engine automatically splits the MIDI velocity (0-127) across the loaded layers. If you load only one, it covers the full range. If you load five, each handles a specific dynamic range for maximum realism.
- **Clear All Layers:** Resets the audio buffers for that specific pad (clean any samples previously loaded on all layers of that specific pad or drum instrument)

8.2. The "Extra Map" Configuration

The Extra Map allows you to expand your kit beyond the 8 standard pads, utilizing the rest of the MIDI keyboard (60 additional notes across 5 octaves).



Mapping a New Articulation:

1. Click the **EXTRA MAP** button.
2. Navigate through the **Octave sub-menus** (Octave 1 to 5).
3. **Reserved Notes:** You will notice some notes are marked as **[RESERVED FOR . . .]**. These are the notes mapped to your main 8 Pads (e.g., C1 for Kick). These cannot be overwritten to prevent trigger conflicts.

4. **Empty Notes:** Click any [Empty] note to open the **Articulation Dialog**. Enter a short name (max 5 characters, e.g., "FLAM" or "COWBL").
5. **Loading Audio to Extra Notes:** Re-open Extra Map, navigate to the note you previously assigned a label to and click on it: only at this point will the window for choosing the samples to assign to that note open. Or better: once a note is named, clicking it again in the menu reveals a specialized sub-menu:
 - **Rename:** Change the label.
 - **Layer 1-3:** Extra articulations support **3 velocity layers** (Low, Mid, High).
 - When a layer of a midi note is already occupied by a sample, the beginning of the name of the sample (i.e. the audio file) will appear, if the layer is still empty the word "Empty" will appear.
 - **Clear All Samples:** Resets the audio buffers for all three layers of that specific note.

8.3. Presets and File Structure (.yadkit)

In versions prior to 1.0.0, the preset format had the .xzkkit extension and was essentially a very simplified version of .sfz. The 1.0.0 version introduces the *.yadkit* format, an XML-based preset file that stores your entire configuration. Since *.yadkit* stores *absolute paths*, moving your sample library to a different drive might require re-loading the samples (they must remain in the same position in which YAD! found them when loading and saving the preset). From version 1.1.0 the internal structure and the logic of the code underlying the preset configuration has changed and the *.yadkit* format searches for the samples in a defined path, i.e. in the "Samples" folder.

- **Saving:** Click **SAVE KIT**. This saves the paths to your audio files, your gain settings, output routing, and all Extra Map data.
- **Loading:** Click **LOAD KIT**. YAD will attempt to rebuild the entire kit.
- **Best Practice:** Keep your drum samples in organized folders. For example, it is recommended to create a folder (let's call it "Library") in which to collect other subfolders, each with the name of your presets; in each of these subfolders you can place both your presets and another subfolder called "Samples" in which to collect the samples related to that preset).

DAW Test Checks:

When loading the plugin into a DAW (Reaper, Ableton, etc.):

1. Multi-Output: Check if the DAW sees the 8 separate outputs. This usually requires "activating" them in the plugin settings within the DAW.
2. Latency: It should be zero (or better yet, equal to the latency setting on your sound card). ASIO drivers are recommended; the standalone supports ASIO, Jack, WDM, WASAPI, WinMM

Technical Implementation Details

For developers or advanced users interested in the "under the hood" solutions:

The Engine: DrumVoice & DrumLayer

The core of the plugin uses a custom-built playback engine. Each drum pad is a DrumVoice containing a dynamic `std::vector` of DrumLayer objects. Each layer holds its own AudioBuffer, allowing for zero-latency switching between velocity zones.

Dynamic Velocity Redistribution

The code employs an automated linear distribution algorithm. When a sample is added or removed, the engine recalculates the `lowVel` and `highVel` integers for every active layer:

$$\text{VelocityRange} = \frac{128}{\text{LoadedLayers}}$$

This ensures the player never encounters "dead zones" in their MIDI performance.

State Management via JSON

The `.xzk` format is based on **JSON (JavaScript Object Notation)**. We utilized `juce::DynamicObject` and `juce::JSON` to create a human-readable save state. This makes the library files lightweight and easy to troubleshoot or manually edit if a file path changes.

Geometry Logic (SUI)

The GUI uses a **Subtractive Bounds** approach. By utilizing `removeFromBottom` and `removeFromLeft`, the interface ensures that the Mixer occupies a fixed 120-pixel safety zone, while the drum kit image scales proportionally to fill the remaining top area, preventing UI "crunching" in small DAW windows.

Technical Insights (Under the Hood)

For those interested in the architecture of YAD, several advanced C++ and JUCE solutions were implemented:

- **Asynchronous UI Pattern:** To prevent DAW freezes, all file browsing and dialog windows (like the naming of extra articulations) use **Modal-less Async Callbacks**. This ensures the audio thread and the GUI thread never block each other.
- **Dynamic Mapping via `std::map`:** Unlike the standard pads which use fixed arrays, the Extra Map utilizes a `[ std::map<int, std::unique_ptr<ExtraArticulation>> ]`. This allows for a memory-efficient "sparse" structure: memory is only allocated for the MIDI notes you actually use.
- **Smart Menu ID Routing:** The popup menu uses a mathematical ID offset system (e.g., IDs 2000+ for layers, 5000+ for clearing). This allows a single callback function to handle hundreds of potential menu items dynamically without complex switch-case structures.

- **XML Serialization:** The `.yadkit` format uses a hierarchical XML structure. This was chosen over binary formats to ensure that presets remain human-readable for debugging and future-proof as the plugin evolves.

SOME DRUM RUDIMENTS

General references to useful drumming techniques for setting up Extra Map

Drum rudiments are foundational sticking patterns—essentially the "alphabet" of drumming—used to develop speed, control, and coordination. There are 40 standard international rudiments, divided into rolls, paradiddles, flams, and drags, which serve as the building blocks for fills and solos. Core rudiments include single/double strokes, paradiddles, and flams.

Please Note: The sequence of letters used to indicate drum rudiments is called sticking. Here are the basic details:

- What it indicates: The sticking specifies which hand should strike the drum, indicating the sequence of strokes (right hand = R, left hand = L).
- Meaning of the letters: R = Right. L = Left. Notational conventions: Uppercase letters (R, L) indicate accented strokes or lead strokes.
- Lowercase letters (r, l) usually indicate softer strokes, such as grace notes in flams or drags (e.g., rL or lR).
-

The use of letters is complemented by the use of numbers.

The sequence of numbers used to indicate drum rudiments (such as the "5-stroke roll," "7-stroke roll," "9-stroke roll," etc.) is generally called **measured rolls or stroke rolls**. These numbers indicate the total number of strokes, including final accents, that make up that specific rudimentary pattern.

Here are the key points:

- What the number indicates: The number specifies how many total strokes make up the roll (e.g., 5-stroke roll = 5 strokes).
- How they are played: They often consist of a series of double strokes (RR LL or LL RR) followed by a final single stroke (R or L).
- The most common: The main measured rolls are the 5th, 7th, 9th, 11th, 13th, 15th, and 17th stroke rolls.
- Context: They are part of the 40 fundamental rudiments codified by the *Percussive Arts Society* (PAS) → <https://pas.org/rudiments/>

Selection from The 40 International Drum Rudiments

According to the [Percussive Arts Society \(PAS\)](#), the 40 standard rudiments are categorized as follows:

I. Roll Rudiments (Single, Double, Triple)

- **Single Stroke Roll:** \(\text{RLRL}\) (alternating). The Single Stroke Roll is the most basic and fundamental **rudiment** of the drums. It consists of a succession of single strokes that constantly alternate between the right hand (R) and the left hand (L) to create a continuous, even, and rapid series of notes.
- **Double Stroke Roll:** A double stroke roll is a fundamental **drum rudiment** consisting of two hits with each hand in an alternating pattern: $\text{RRLLcap R cap R cap L cap L}$ (RRLL). It allows drummers to create a fast, continuous, and articulate roll sound by using a combination of wrist strokes and the stick's natural rebound, aiming for even volume between all strokes.
- **Multiple Bounce Roll:** A Multiple Bounce Roll - often called a **buzz roll**, press roll, or orchestral roll—is a drum rudiment where each stick stroke produces multiple, rapid bounces (buzzes) rather than a single or double rebound. It creates a smooth, continuous, and sustained sound commonly used in orchestral, concert band, and jazz music.

II. Paradiddle Rudiments

- **Single Paradiddle:** The single paradiddle is the first of the four basic paradiddle rudiments, (which also include double paradiddles, triple paradiddles, and paradiddle-diddles) consisting of two alternating single strokes followed by a double stroke (R L R R or L R L L). It combines single and double strokes to naturally alternate leading hands, making it essential for playing around the drum set, creating accents, and building control.

Key Aspects of the Single Paradiddle:

- **Sticking Pattern:** $\text{RLRRcap R cap L cap R cap R}$ ($\text{\(RLRR\)}$)
or $\text{LRLLCap L cap R cap L cap L}$ ($\text{\(LRLLCap\)}$).
 - **Composition:** It mixes single strokes ("para") and a double stroke ("diddle").
 - **Accents:** It is commonly played with an accent on the first note of each group (the first single stroke).
 - **Purpose:** It helps drummers switch lead hands effortlessly, facilitates movement around the kit, and is used extensively in beats and fills.
- **Double Paradiddle:** A double paradiddle is a 6-note drum rudiment consisting of four alternating single strokes followed by a double stroke (RLRLRR or LRLRLL). It is a variation of the single paradiddle, adding an extra pair of alternating strokes ("para-para") before the "diddle" (double), and is often used to build speed, endurance, and coordination.

Key Aspects of the Double Paradiddle:

- **Sticking:** $\text{RLRLRRcap R cap L cap R cap L cap R cap R}$ (RLRLRR)
 $\text{LRLRLLcap L cap R cap L cap R cap L cap L}$ (LRLRLL)
- **Rhythm:** Primarily used in triplets or 6/8 and 12/8 time signatures, making it great for jazz, Latin, and triplet-feel fills.

- **Accent:** The traditional method is to accent the very first note of each 6-note grouping, although practitioners may accent the first two single strokes for a "para-para-paradiddle" feel.
- **Triple Paradiddle:** A triple paradiddle is a 8-note drum rudiment consisting of three sets of alternating single strokes followed by a double stroke (diddle), usually played as Rlrlrlrr Lrlrlrl. It is a natural progression from single/double paradiddles, commonly accented on the first note to create a flowing, alternating rhythm

Key Components:

- **Sticking Pattern:** RLRLRLRRcap R cap L cap R cap L cap R cap L cap R cap R (*RLRLRLRR*)
- **Structure:** It consists of six alternating single strokes, followed by one double stroke (6 singles + 1 double = 8 notes).
- **Accents:** Traditionally, the first note of each eight-note grouping is accented, creating a "down-tap-tap-tap-tap-tap-up-tap" dynamic flow.
- **Purpose:** It is used to develop stick control, alternating hand-to-hand, and is excellent for building speed in 16th or triplet notes.
- **Mnemonic:** A common way to remember the pattern is "Par-a-par-a-par-a-diddle" or "I'm a triple, you're a triple"
- **Single Paradiddle-diddle:** is a six-note drum rudiment (number 19 of the 40 P.A.S. rudiments) consisting of a single stroke alternation followed by two double strokes (diddles). It is characterized by its **RLRRLL** or **LRLRRR** sticking, creating a rapid, rolling feel often used in triplet or 6/8 time signatures

Key Aspects of the Single Paradiddle-Diddle:

- **Sticking Pattern:** Right-Left-Right-Right-Left-Left (RLRRLL) or Left-Right-Left-Left-Right-Right (LRLRRR). Unlike the single paradiddle (*(RLRR \ LRLR)*), the paradiddle-diddle adds an extra "diddle" (double stroke) to create a six-note pattern rather than four.
- **Rhythmic Feel:** It has a triplet feel and is commonly played over 6 notes.
- **Usage:** The paradiddle-diddle is excellent for moving around the drum kit, creating fast fills, and developing speed because the pattern repeats without needing to immediately alternate the leading hand, unlike the basic single paradiddle.
- **Accents:** Commonly practiced by accenting the first note (downbeat) of the sequence: **Rlrrll Lrlrrr**.

III. Flam Rudiments

- **Flam:** consisting of two notes played in close succession with both sticks: a soft, low "grace note" followed immediately by a louder "primary" (or accented) stroke. It is used to thicken or accent a single beat, creating a richer sound than a single stick hit; frequently used in various styles including rock, jazz, and marching, and serves as the basis for other rudiments such as the flam accent, flam tap, and flamacue

Key aspects of a flam include:

- **Sticking Pattern:** I R l cap R l R or r L r cap L r L (A low grace note followed by a high primary note).
 - **Structure:** It consists of a ghost note followed by an accented note.
 - **Sound:** It produces a "fuller" or "thicker" sound on a single note.
 - **Notation:** In written music, it appears as a small "grace note" immediately preceding a regular note.
 - **Naming:** The flam is named based on the hand that plays the primary (loud) note.
- **Flam Accent:** combines a single stroke roll with a flam, played in a triplet rhythm. It consists of three notes—accented flam, tap, tap—alternating hands (e.g., \ (r L \ R \ L \ I R \ L \ R \)), often used for creating textured drumming fills and patterns with a "flam-tap-tap" sound.

Key aspects of a flam include:

- **Sticking Pattern:** I R L r L R l cap R cap L r cap L cap R (l R L r L R)
 - **Structure:** The pattern is usually played as a triplet: **Flam** (accented) - Tap – Tap.
 - **Common Use:** Flam accents are excellent for drum fills and solos, often described as a "single stroke roll with a flam on the downbeat".
 - **Key Tip:** Keep the grace notes low and avoid "flat flams" (where both sticks hit simultaneously) to maintain a crisp sound.
- **Flam Tap:** A flam tap combines a flam with a tap, essentially acting as a double-stroke roll where the first note of each pair is replaced by a flam. It alternates between hands (right-left-left, left-right-right) and is used to add a "thicker" sound to a pattern. The primary note of the flam and the tap are usually accented, creating a loud-soft-loud feel or a "double-accent" effect.

Key Aspects of the Flam Tap:

- **Sticking Pattern:** I R R l cap R cap R (l R R) > r L L r cap L cap L (r L L)
 - **Structure:** It consists of a flam (a grace note followed by a primary note) followed by a single tap, alternating hands: \ (r L \ L \ I R \ R \).
 - **Sticking:** Right-hand flam followed by a right tap (\ (r L \ L \)), then a left-hand flam followed by a left tap (\ (I R \ R \)).
 - **Purpose:** Excellent for warming up hands and improving dexterity, often utilized in drumming to add flair to drum fills
- **Flamacue:** combines a flam at the beginning and end of a five-stroke pattern, characterized by a distinct accent on the second stroke. It is often described as a single-stroke five with added grace notes, featuring a syncopated feel that shifts emphasis away from the downbeat

Key Aspects of the Flamacue:

- **Sticking:** I R L R R l cap R cap L cap R cap R - (I R L R L r R vs r L R L R I L)
- **Structure:** It consists of five notes, often structured as a flam, an accent, two taps, and a final flam (e.g., I R L R L I R or r L R L R r L).
- **Accent:** The crucial accent is on the **second stroke** (the 'e' of the count), rather than just the first stroke.
- **Sound/Feel:** It creates a lilting, syncopated cadence often found in marching and rudimental drumming.
- **Application:** Useful for drum fills and adding rhythmic interest to grooves.

- **Flam Paradiddle:** is a hybrid drum rudiment that combines a single paradiddle with a flam on the first beat of the sequence. It acts as an accented, embellished version of a standard paradiddle, following the pattern: Flam-Right, Left, Right-Right / Flam-Left, Right, Left-Left (flam L R L L)

Key Characteristics:

- **Structure:** It consists of 6 strokes per alternating sequence (or 4 main movements). The structure is: [Flam] [Tap] [Tap] [Tap] [Flam] [Tap] [Tap] [Tap] (with internal doubles).
 - **Sticking:** The right-handed version is generally: IRLRR rLRLl (lowercase indicates the grace note). IRLRRLl cap R cap L cap R cap R cap L (*lRLRRL*)
rLRLLEr cap L cap R cap L cap L cap E (*rLRLLE*)
 - **Sound:** Often called a "flamidal", this rudiment adds a thicker, more textured sound to standard paradiddle sticking, perfect for drum solos and fills.
 - **Use Cases:** It is excellent for moving around the drum kit, particularly for accenting toms or cymbals while maintaining a steady snare rhythm
- **Pataflafla:** consisting of a single stroke roll with a flam played on the first and last notes of a four-note sequence. It is known for its distinctive sound (flam-tap-tap-flam)

Key Aspects of the Pataflafla:

- **Sticking Pattern:** The basic pattern is: [Flam] - Tap - Tap - [Flam] (e.g., right-hand flam, left tap, right tap, left-hand flam).
IRLLRrLRRLl cap R cap L cap L cap R r cap L cap R cap R cap L (*lRLLRrLRRL*)
 - **Sounds Like It Sounds:** The name is onomatopoeic, derived from the French phrase describing the rhythm (it has strong historical roots in French military music).
 - **Usage:** It is great for drum fills and, because it's played on the snare, it can be applied to the hi-hat or ride cymbals as well
- **Swiss Army Triplet:** combines a flam with a double-stroke roll played within a triplet rhythm. It consists of three notes: a flam followed by two taps, typically played with a "right-right-left" (rR L) or "left-left-right" (lL R) pattern, creating a flowing, bouncy, and rhythmic feel used frequently in drum fills.

Key Features:

- **Sticking Pattern:** The basic pattern is **rR L rR L** (right-hand lead) or **lL R lL R** (left-hand lead). IRLRl cap R cap L cap R (*lRLR*)
- **Structure:** It is essentially a double-stroke roll with a flam on the first note, followed by a single tap from the opposite hand.
- **Application:** Commonly used in marching percussion (tenor drums) and on drum sets for rolls, solos, and creative fills.

IV. Drag Rudiments

- **Drag:** consisting of two quick, quiet grace notes (a "diddle") played immediately before a primary, accented note. It functions as a "double followed by a single" (lL R or rR L), acting as a musical embellishment that adds texture and flare to beats and fills
- **Key Aspects of the Drag Rudiment:**
 - **Structure:** It is similar to a flam, but uses two grace notes instead of one.

- **Sticking:** Two quick grace notes: *llRl l cap R (llR)* or *rrLr r cap L (rrL)*. It alternates, starting with two soft left-hand notes leading to a right-hand accent (*\(llR\)*), then two soft right-hand notes leading to a left-hand accent (*\(rrL\)*).
- **Execution:** The grace notes should be played low to the drumhead and very tight (close) to the main stroke to create the signature "dragging" sound.

Variations of Drag:

- **Open Drag:** The two grace notes are distinct and played as a fast double-stroke.
- **Closed/Buzz Drag:** The grace notes are played as a quick buzz/multiple bounce stroke.
- **Single Drag Tap:** combines a soft two-note drag (grace notes) with a primary accented tap. It creates a distinct "drag-tap" sound pattern, typically played as *llR l* or *rrL r*, alternating hands to add emphasis and texture to drumming rudiments, fills, and solo patterns.

Key Features & Techniques

- **Structure:** A drag (a small, unaccented double stroke) followed by an accented tap.
- **Sticking:** Right-hand drag, left-hand accent (*llR l*) or Left-hand drag, right-hand accent (*rrL r*).
llRLRl l cap R cap L cap R (llRLR) > rrLRLr r cap L cap R cap L (rrLRL)
- **Rhythm:** Can be played in duple time (*\(1\ \&\ 2\ \&\dots\)*) or triplet time (*\(1-2-3,\ 1-2-3\)*).
- **Common Applications:** Often used in drum corps, quick-step marches, and for enhancing drum kit fills
- **Double Drag Tap:** featuring two drags followed by an accented tap, providing a dense, rhythmic "roll" feel often used in 12/8 time or with a triplet feel. It consists of alternating hands, with the basic pattern being: two small drags followed by an accented tap (e.g., *llR llR L* or *rrL rrL R*). Key elements of this rudiment include:
 - **Structure:** It is a variation of the single drag tap but features an extra drag for more decoration and flair.
 - **Sticking Pattern:** The pattern uses a double drag followed by a tap, often played as *"llR llR L"* (left hand/left hand/right hand accent) or *"rrL rrL R"*.
llRllRLRl l cap R l l cap R cap L cap R (llRllRLR).
 - **Rhythmic Feel:** It has a distinct "triplet" or "rolling" feel and is sometimes referred to as an "offset triple stroke roll with drags".
 - **Application:** It is commonly played in 3/4 or 12/8 time, often with the drags on a different surface (like a hi-hat) than the accents (on the snare)
- **Drag Paradiddle #1 & #2:** A Drag Paradiddle is a hybrid drum rudiment combining a DRAG (two grace notes) with a PARADIDDLE (*RLRR LRLl*). It adds a drag to the beginning of a paradiddle to create a more complex, articulated pattern. It is commonly

used to enhance technical control and adds a unique, "dragged" sound to fills or snare work

- **Ratamacues:**

- **Single/Double/Triple Ratamacue:** Ratamacues are triplet-based drag rudiments, characterized by a "drag-drag-drag" (or similar) pattern followed by a four-stroke sequence with an accent on the last note. They are often played in jazz, Latin, and marching styles.
- **Single Ratamacue:** A 4-stroke rough with a drag, played *drag-left-right-left* (accent on the last note).
- **Double Ratamacue:** Starts with two consecutive drags followed by the single ratamacue pattern.
- **Triple Ratamacue:** Consists of three consecutive drags (drag-drag-drag) followed by the single ratamacue pattern.

Key Aspects:

- **Sticking (Triple):** Often played as $\backslash(IIR\ IIR\ IIR\ LRL)$ or $\backslash(rrL\ rrL\ rrL\ RLR)$.
- **Accent:** The primary accent falls on the final stroke.
- **Feel:** They provide a rolling triplet feel, commonly used in marching snare or as drum kit fills.

